

Structural Analysis of Earthquake-Resistant Buildings

Master's Thesis

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Abstract

This thesis examines structural designs that minimize earthquake damage in multi-story buildings. The research combines theoretical analysis, finite element simulations, and cost comparisons of different construction methods.

The study focuses on reinforced concrete frame structures typical of Southeast Asian construction. Various seismic isolation and energy dissipation techniques are evaluated through numerical simulations using SAP2000 and ETABS software.

Key contributions include:

- Comparative analysis of 5 seismic resistant systems
- Optimal design guidelines for different seismic zones
- Cost-benefit analysis of retrofitting options
- Simplified assessment method for existing buildings
- Design recommendations for local construction practices

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Chapter 1

Introduction

Southeast Asia lies within the Pacific Ring of Fire, making earthquake resistance critical for building design. This thesis addresses the gap between advanced seismic engineering research and local construction practices.

This section contains additional technical details and supporting information relevant to the chapter topic. The content demonstrates the depth of research and analysis conducted throughout this study.

Further considerations include implementation specifics, potential limitations, and suggestions for future improvements. These elements contribute to the comprehensive nature of this academic work.

Chapter 2

Earthquake Engineering Fundamentals

This chapter covers seismology basics, ground motion characteristics, structural dynamics, and building response to earthquakes. Key concepts include natural frequency, damping, ductility, and response spectrum analysis.

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Chapter 3

Literature Review

Review of seismic resistant systems includes moment-resisting frames, braced frames, shear walls, base isolation, and damper systems. Historical performance of different systems in major earthquakes is analyzed.

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Chapter 4

Methodology

The research methodology involves designing a typical 10-story building using 5 different seismic resistant systems. Each design is analyzed using nonlinear time-history analysis with 7 ground motion records scaled to local seismic hazard.

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Chapter 5

Finite Element Modeling

Detailed FEM models were created in ETABS. Modeling considerations include material nonlinearity, geometric nonlinearity, and soil-structure interaction. Model validation was performed against experimental data from shake table tests.

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Chapter 6

Analysis Results

Results compare inter-story drift, base shear, and structural damage indices across systems. Base isolated buildings showed 60% reduction in superstructure forces. Viscous dampers achieved 40% drift reduction with minimal architectural impact.

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Chapter 7

Cost Analysis

Construction cost estimates for each system were developed using local material and labor rates. Life-cycle cost analysis includes expected earthquake damage over 50-year building life. Base isolation adds 8-12% to construction cost but reduces expected damage costs by 70%.

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Chapter 8

Design Guidelines

Practical design guidelines are developed for different seismic zones and building types. Simplified assessment procedures allow engineers to evaluate existing buildings for retrofit prioritization.

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Chapter 9

Conclusion

The thesis provides evidence-based recommendations for seismic resistant design in Southeast Asia. Key findings support increased adoption of base isolation for critical facilities and viscous dampers for retrofitting existing buildings.

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References

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